RISK PORTFOLIO REPORTING

Automated risk portfolio reporting

Muster-Sparkasse Nord (fiktiv)

Reporting date 2026-09-30 | Peer group PG_URBAN

Synthetic demonstration report

This report combines a shared ingestion and data-quality platform with two analytical modules: Rating & Scoring and Loss Estimation. It is a runnable prototype built exclusively on fictional Sparkassen, portfolios and cashflows.

Institution scope	**Muster-Sparkasse Nord (fiktiv)**	Community view or single fictional institution.
Current EAD	**685,934,216 EUR**	Positive exposure at default in the reporting snapshot.
Exposure-weighted PD 12M	**1.48%**	Configured demonstration PD mapping, not a validated model.
Observed 12M default rate	**2.42%**	Backward-looking 12-month cohort.
Observed 12M recovery rate	**21.38%**	Realised recovery cashflows through the reporting date.
Observed 12M loss rate	**78.62%**	EAD at default minus realised recoveries.
Data quality score	**99.81%**	Valid records / received records across completed batches.
Data version	**RUN-20261111005418-7B456F2D**	Latest pipeline run identifier.

SYNTHETIC DEMONSTRATION DATA - NOT REGULATORY REPORTING AND NOT REAL SPARKASSEN OR CUSTOMER INFORMATION

No real Sparkasse, customer, employee or portfolio is represented. Values are generated with a fixed seed and controlled data-quality defects for demonstration. The report is not regulatory reporting and must not be used for supervisory, accounting or customer decisions.

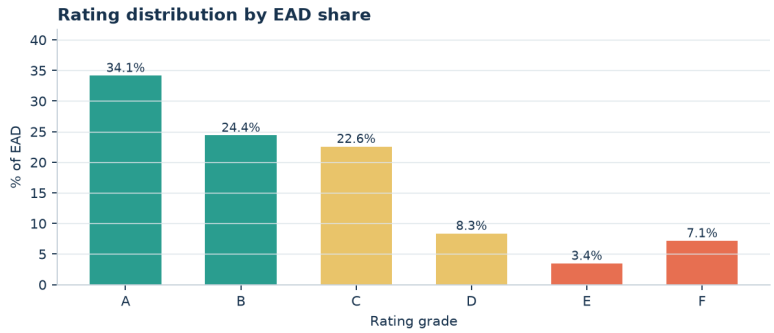
1. Executive overview

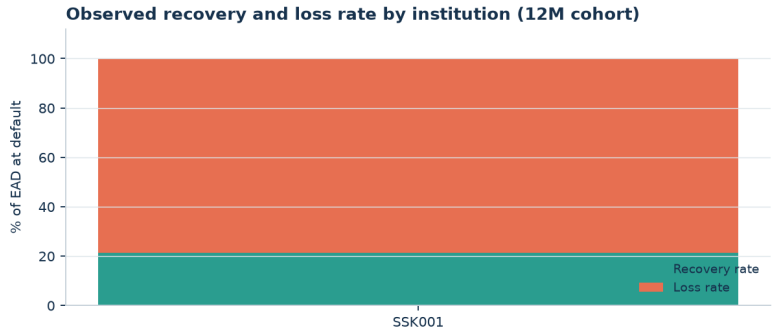
One reporting date, two analytical modules and a shared data foundation.

At 2026-09-30, the selected scope contains 449 exposures with total EAD of EUR 685,934,216. The exposure-weighted configured PD 12M is 1.48%. The observed 12-month default rate for the backward cohort is 2.42% (9 defaults on a base of 372 performing exposures). Observed recovery through the reporting date is 21.38%, giving an observed loss rate of 78.62% on EAD at default of EUR 7,829,957. Collateral coverage of current EAD is 42.62%; the secured EAD share is 42.60%.

Data quality is 99.81% record pass rate across 4 completed batches. 16 error findings and 0 warning findings were recorded; 15 records are retained in quarantine and were not loaded to curated tables. Warnings remain in curated data with their rule evidence, rather than being silently deleted.

Portfolio PD 12M	**1.48%**	Exposure-weighted configured PD.
Observed 12M default rate	**2.42%**	Count-based backward cohort.
Investment-grade share	**58.52%**	EAD rated A or B.
Upgrade / downgrade rate	**14.21% / 14.21%**	Matched 12-month migration.
Recovery rate	**21.38%**	Recoveries / EAD at default.
Loss rate	**78.62%**	Loss / EAD at default.
Collateral coverage	**42.62%**	Haircut-adjusted collateral / current EAD.
Quality score	**99.81%**	Valid / received records.



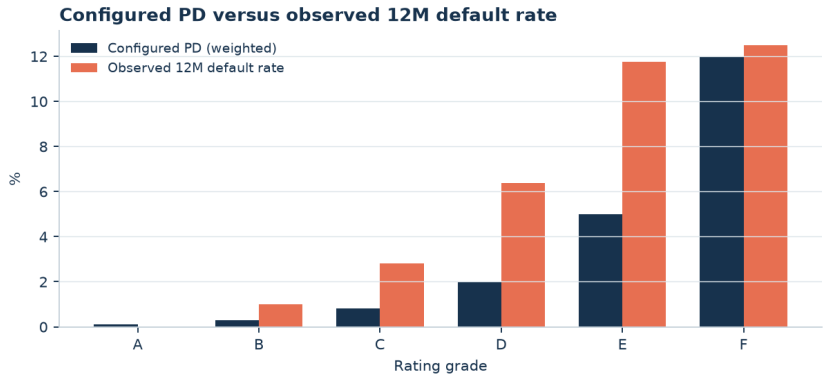


2. Rating & Scoring

Distribution, configured PD 12M, observed default rates and rating migration.

2.1 Rating distribution

Rating	Exposures	EAD (EUR)	Share count	Share EAD	PD 12M	Expected default EAD
A	150	234,217,139	33.41%	34.15%	0.10%	234,217
B	107	167,212,547	23.83%	24.38%	0.30%	501,638
C	89	154,685,096	19.82%	22.55%	0.80%	1,237,481
D	59	57,232,731	13.14%	8.34%	2.00%	1,144,655
E	28	23,642,244	6.24%	3.45%	5.00%	1,182,112
F	16	48,944,459	3.56%	7.14%	12.00%	5,873,335



2.2 Observed default rates

The observed default rate uses a 12-month backward cohort. The denominator is the set of performing exposures in the latest quarter snapshot on or before reporting date minus 12 months. A default is counted once per exposure when the first observed default date lies after that observation start and on or before the reporting date. The configured PD is shown separately and is not calibrated to the observed rate in this prototype.

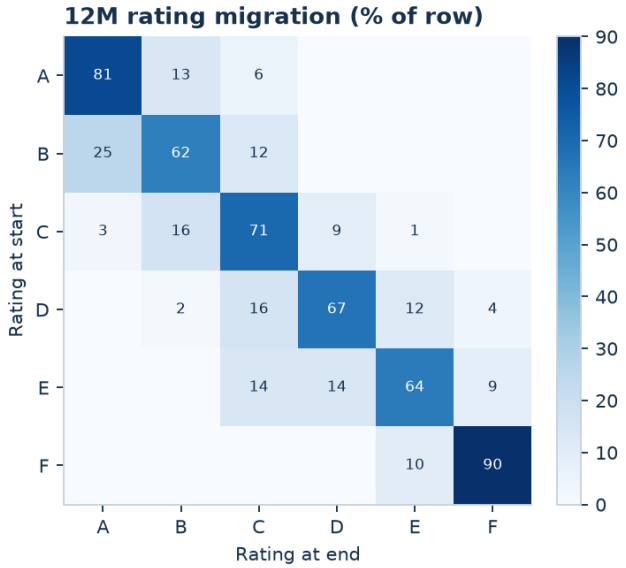
Rating	Base exposures	Base EAD	Defaults	Default EAD	Observed rate	Observed rate (EAD)	Configured PD
A	129	218,126,717	0	0	0.00%	0.00%	0.10%
B	100	174,843,813	1	703,991	1.00%	0.40%	0.30%
C	71	88,443,756	2	753,233	2.82%	0.85%	0.80%
D	47	47,433,875	3	1,017,747	6.38%	2.15%	2.00%
E	17	34,584,277	2	524,362	11.76%	1.52%	5.00%
F	8	28,310,754	1	822,210	12.50%	2.90%	12.00%

Bank	Base exposures	Defaults	Observed rate	Observed EAD rate	Configured PD
SSK001	372	9	2.42%	0.65%	1.27%

2.3 Rating migration

Migration compares 2025-09-30 with 2026-09-30. 373 exposures are matched. 77 entries and 16 exits are reported separately. Upgrade rate is 14.21%; downgrade rate is 14.21%.

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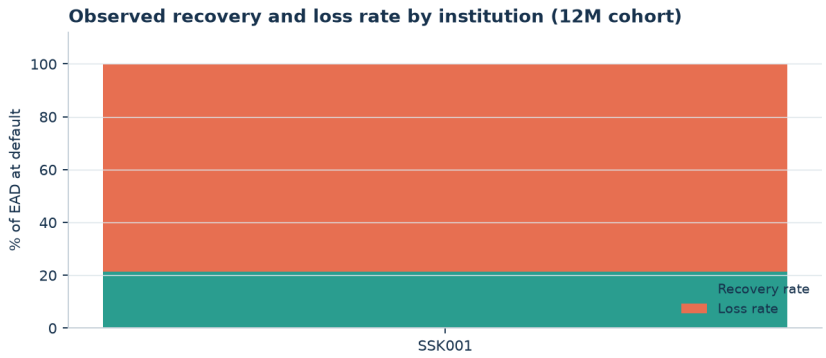
From / to	A	B	C	D	E	F
	80.95%	13.49%	5.56%	0.00%	0.00%	0.00%
	25.00%	62.50%	12.50%	0.00%	0.00%	0.00%
	2.94%	16.18%	70.59%	8.82%	1.47%	0.00%
	0.00%	1.96%	15.69%	66.67%	11.76%	3.92%
	0.00%	0.00%	13.64%	13.64%	63.64%	9.09%
	0.00%	0.00%	0.00%	0.00%	10.00%	90.00%

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3. Loss estimation

Observed recovery, loss severity, collateral coverage and cashflow timing.

The observed default cohort covers 10 defaults with EAD at default of EUR 7,829,957. The observed loss amount is EUR 6,155,849. Collateral is used only for coverage analysis and is not subtracted again from loss, preventing double counting. Observed recovery rate is 21.38%; observed loss rate is 78.62%. 30.00% of defaults have no observed recovery by the reporting date.

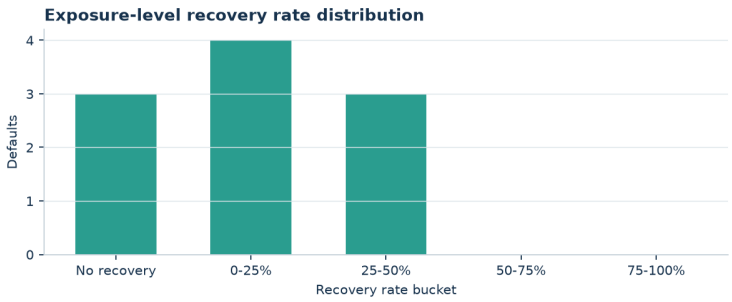


3.1 Recovery and loss by institution

Bank	Defaults	EAD at default	Recoveries	Recovery rate	Loss	Loss rate
SSK001	10	7,829,957	1,674,108	21.38%	6,155,849	78.62%

3.2 Recovery rate by rating at default

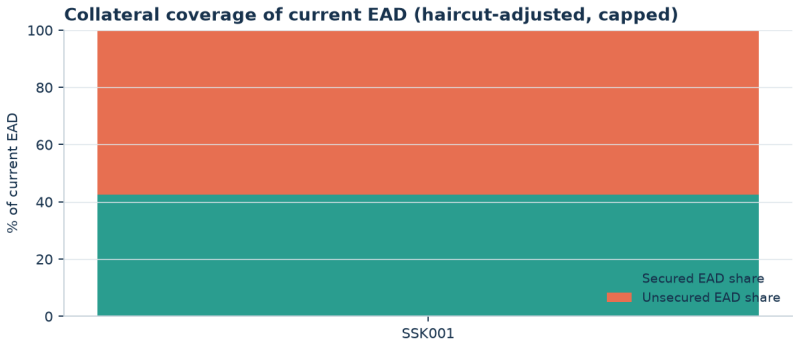
Rating	Defaults	EAD at default	Recoveries	Recovery rate	Loss rate
A	1	3,992,107	900,433	22.56%	77.44%
B	1	703,991	264,735	37.60%	62.40%
C	2	753,233	293,362	38.95%	61.05%
D	3	1,019,733	215,578	21.14%	78.86%
E	2	530,105	0	0.00%	100.00%
F	1	830,787	0	0.00%	100.00%



4. Collateral and recovery cashflows

Latest valuations, haircuts, coverage, cashflow timing and cumulative recovery.

Collateral coverage is based on the latest valuation on or before the reporting date. Configured haircuts are 20% for real estate, 5% for financial collateral and 40% for other collateral. Total adjusted collateral is EUR 292,359,279; the unsecured EAD amount is EUR 393,693,923. 0 latest collateral items exceed the 3-year valuation age warning threshold.



4.1 Collateral by institution

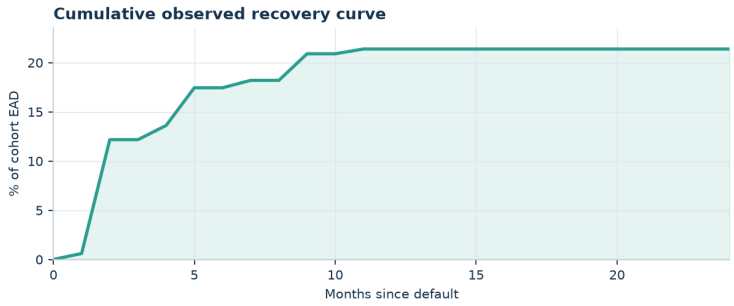
Bank	Exposures	Current EAD	Collateral value	Adjusted value	Secured EAD share	Coverage
SSK001	449	685,934,216	n/a	292,359,279	42.60%	42.62%

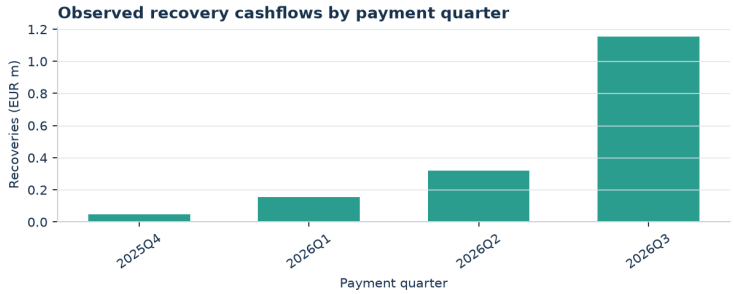
4.2 Collateral type mix

Type	Items	Value	Adjusted	Haircut reduction	Share adjusted
REAL_ESTATE	204	281,184,076	224,947,261	56,236,815	76.94%
FINANCIAL_COLLATERAL	47	47,288,177	44,923,768	2,364,409	15.37%
OTHER	38	37,480,418	22,488,251	14,992,167	7.69%

4.3 Recovery cashflow timing

Payment period	Cashflows	Recovered	Share	Cumulative share
2025Q4	1	46,863	2.80%	2.80%
2026Q1	1	154,429	9.22%	12.02%
2026Q2	4	317,478	18.96%	30.99%
2026Q3	6	1,155,338	69.01%	100.00%



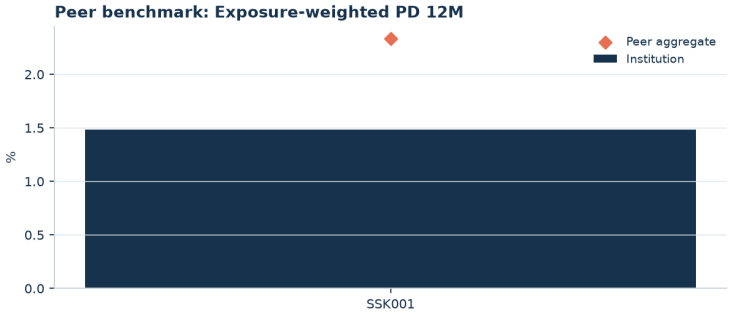


5. Peer benchmarking

Numerator/denominator peer aggregation, excluding the institution from its own peer reference.

Peer values are calculated from summed numerators and denominators, not by averaging bank-level ratios. The bank under review is excluded from its own peer group. Relative differences above 10% are labelled BETTER or WORSE when the metric has an unambiguous direction; neutral rating-share metrics are informational.

Metric	Bank	Peer aggregate	Gap	Rank	Flag
Adjusted collateral coverage of current EAD	42.62%	26.39%	16.23%	3	BETTER
Observed 12M default rate (count)	2.42%	8.14%	-5.72%	3	BETTER
Observed 12M loss rate (EAD-weighted)	78.62%	83.57%	-4.95%	3	IN_LINE
Exposure-weighted PD 12M	1.48%	2.33%	-0.85%	3	BETTER
Observed 12M recovery rate (EAD-weighted)	21.38%	16.43%	4.95%	3	BETTER

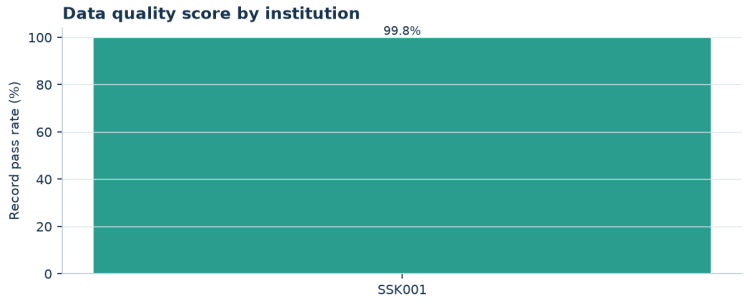




6. Data quality and lineage

Validated records, error quarantine, warnings and pipeline traceability.

The data quality score is 99.81% (valid records / received records). 4 batches completed and 0 duplicate batches were skipped. 16 error findings and 0 warning findings are available in the audit tables. Quarantine holds 15 records; these were not loaded into curated analytical tables. Warnings are retained in curated data with their rule evidence.



6.1 Quality by institution

Bank	Batches	Received	Valid	Rejected	Score	Errors	Warnings
SSK001	4	7,722	7,707	15	9981.00%	16	0

6.2 Top data-quality rules

Rule	Name	Severity	Findings	Records
COMP-EXP-002	Bank ID is required	ERROR	1	1
COMP-EXP-003	Reporting snapshot date is required	ERROR	1	1
COMP-EXP-005	Rating grade is required	ERROR	1	1
CONS-COL-003	Collateral exposure exists	ERROR	1	1
CONS-EXP-002	Rating grade is on the configured scale	ERROR	1	1
CONS-RAT-002	Rating history grade is valid	ERROR	1	1
CONS-RAT-003	Rating history exposure exists	ERROR	1	1
CONS-REC-002	Recovery exposure exists	ERROR	1	1
PLAUS-COL-001	Collateral value is non-negative	ERROR	1	1
PLAUS-EXP-001	EAD is non-negative	ERROR	1	1
PLAUS-REC-001	Recovery amount is non-negative	ERROR	1	1
UNIQ-COL-001	Collateral valuation is unique within batch	ERROR	1	1

6.3 Data lineage and batch audit

Bank	Dataset	Batches	Received	Valid	Rejected	Latest import
SSK001	collateral	1	684	681	3	2026-11-11
SSK001	exposures	1	3,468	3,462	6	2026-11-11
SSK001	rating_history	1	3,469	3,466	3	2026-11-11
SSK001	recovery_cashflows	1	101	98	3	2026-11-11

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7. Methodology and assumptions

Definitions that are deliberately explicit because the prototype is a demonstration.

7.1 Shared data platform

Heterogeneous CSV, Excel and source-database files are mapped into canonical staging entities. Completeness, uniqueness, consistency and plausibility rules run before curated loading. Error records are quarantined with their rule IDs and raw JSON; warnings are retained in curated data. Every batch, quality finding and pipeline run is traceable to a source file hash and import timestamp.

7.2 Rating & Scoring

Rating distribution and expected default are based on the configured, explicitly fictional rating-to-PD mapping: A 0.1%, B 0.3%, C 0.8%, D 2.0%, E 5.0%, F 12.0%. 12-month through-the-cycle-style demonstration PD by rating grade. Values are configurable assumptions, not validated model parameters. Observed default rates use a separate 12-month backward cohort and are not used to calibrate the configured PD in this prototype. Migration compares two snapshots, reports entries and exits separately and uses matched exposures for the transition matrix.

7.3 Loss Estimation

EAD at default is the latest available exposure snapshot on or before the observed default date. Recoveries are cashflows paid after default and on or before the reporting date. Loss is EAD at default minus realised recoveries. No discounting, indirect workout costs or collateral double counting are applied. Collateral coverage uses the latest valuation on or before the reporting date and configured haircuts REAL_ESTATE 20%, FINANCIAL_COLLATERAL 5%, OTHER 40%. Coverage is shown for context, not subtracted again from loss.

7.4 Benchmarking

Benchmark values are calculated from summed numerators and denominators across peer institutions. The institution under review is excluded from its own peer reference. Configured peer groups: PG_URBAN = SSK001, SSK003, SSK005; PG_REGIONAL = SSK002, SSK004. A small peer group means the peer reference is illustrative rather than statistically robust.

7.5 Data quality score

The quality score is the record pass rate: 100 x valid records / received records across completed batches. It is not a measure of model accuracy, calibration or regulatory fitness. The score is computed per batch and aggregated by totals, not by averaging batch percentages.

7.6 Limitations

This is a local DuckDB prototype with synthetic data. It has no real external integrations, no production identity management, no regulatory sign-off and no validated credit-risk models. Peer groups are tiny and illustrative. The configured PDs, haircuts and loss assumptions are demonstration parameters only. A production implementation would require model validation, data-lineage controls, access management, reconciliation, backtesting, audit sign-off and an enterprise database/BI deployment.

7.7 Reproducibility

Synthetic data is generated with a fixed seed and a generation manifest. The pipeline can be rerun with python synthetic_data/generate_data.py followed by python -m etl.pipeline --reset-db. Reports are logged in the report_log table and distribution is simulated locally.

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Appendix A. Report audit

Generated artefact metadata and disclaimer.

Item	Value
Report title	Risk Portfolio Reporting - 2026-09-30
Organisation label	Fictional Sparkassen Reporting Community
Scope	Muster-Sparkasse Nord (fiktiv)
Peer group	PG_URBAN
Generated at	2026-11-11 00:57:17
Pipeline run	RUN-20261111005418-7B456F2D
Batches completed	20
Rows received	36003
Rows valid	35924
Rows rejected	79
Disclaimer	SYNTHETIC DEMONSTRATION DATA - NOT REGULATORY REPORTING AND NOT REAL SPARKASSEN OR CUSTOMER INFORMATION

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The report is generated from synthetic demonstration data. It must not be represented as a real Sparkasse report, a regulatory submission, a validated model output or a customer communication.